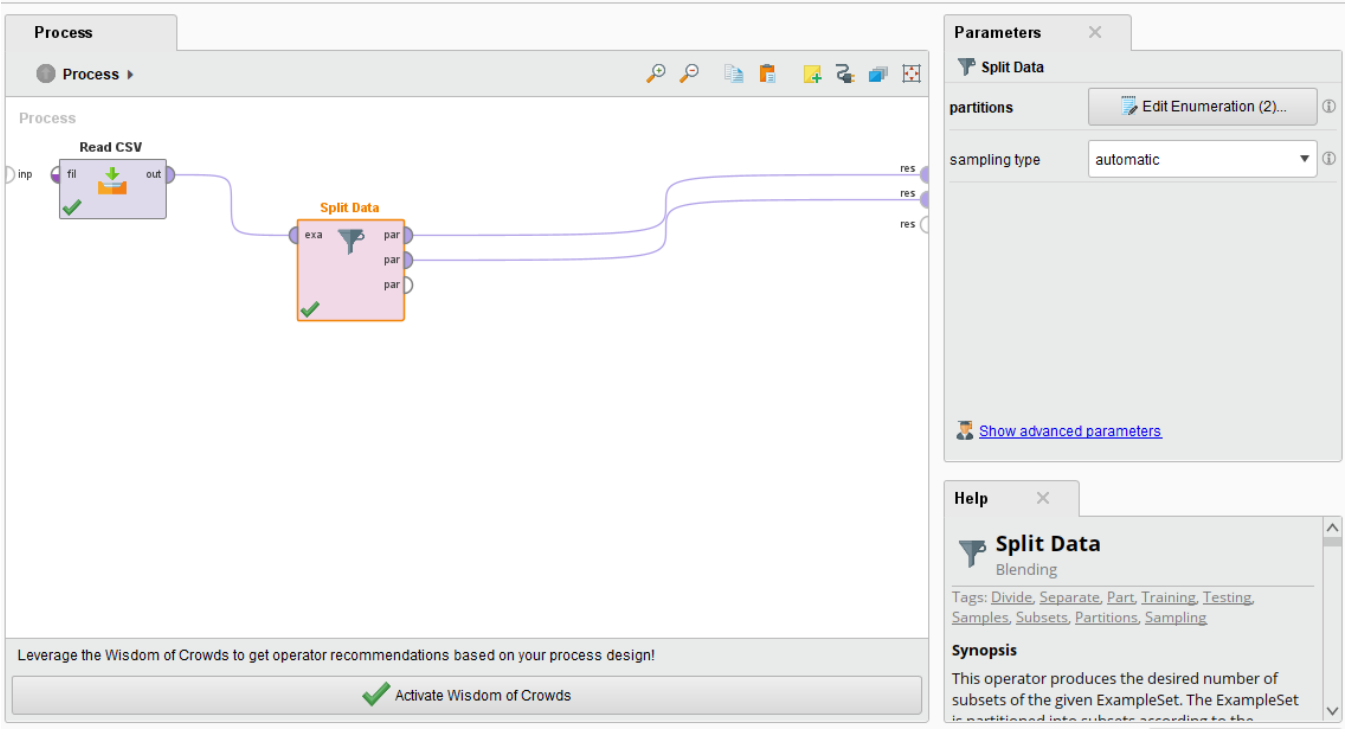
Predicting Virat Kohli's Runs in Altair AI Studio

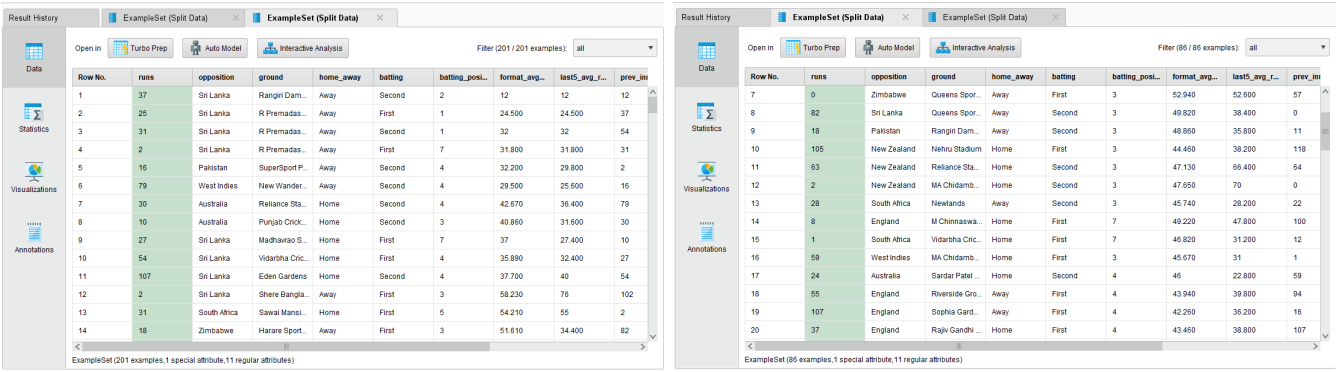
Goal: predict the **runs** column (how many runs Virat Kohli scores in an ODI innings) from the other (known) columns such as opposition, ground, home_away, batting, batting_position, format_avg_before, last5_avg_runs and prev_innings_runs. Steps: split the data, then build a Decision Tree.

Step 1: Split the Data

The data was imported with **Read CSV** and connected to **Split Data**. Its parameters use **automatic sampling** and two partitions set with Edit Enumeration, which divide the matches into about **70% training data (201 matches)** and **30% testing data (86 matches)**. In both parts, **runs** is the special attribute (the label, what we predict) and the other 11 columns are regular attributes.



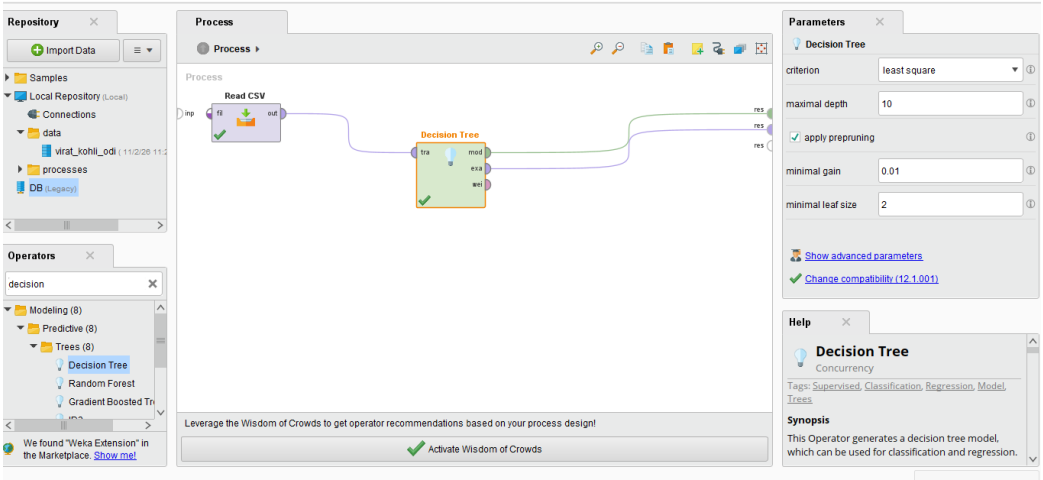
Split Data design and its parameters.



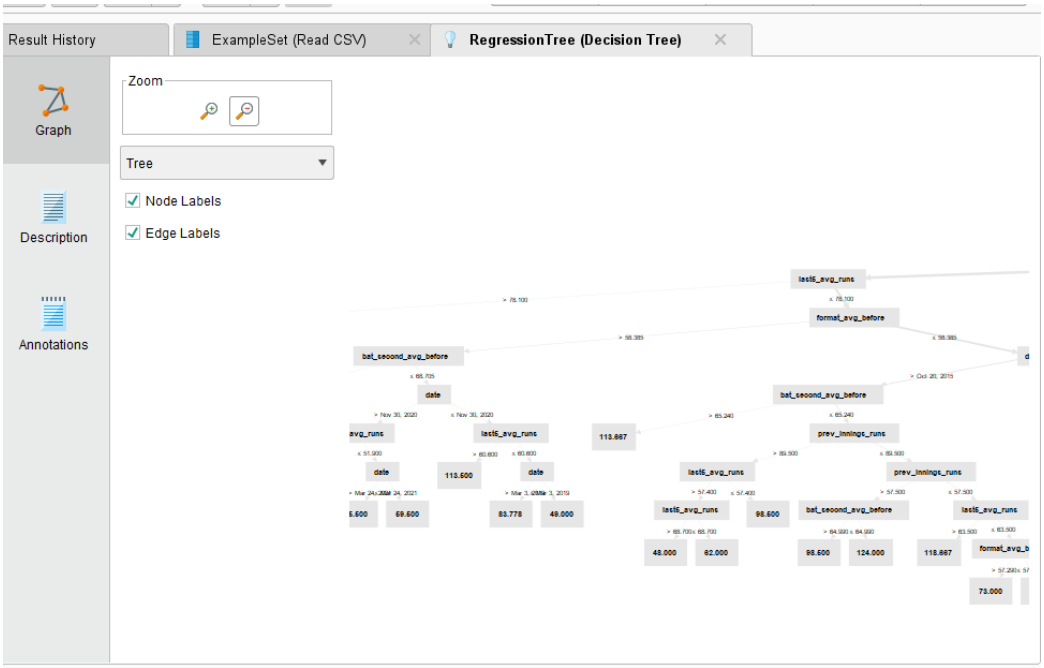
Left: training data (201 examples). Right: testing data (86 examples).

Step 2: Decision Tree

The data from **Read CSV** is connected to the **Decision Tree** operator, and its model output is sent to the results. Because **runs** is a numeric label, AI Studio builds a regression tree. The operator uses the **least square** criterion, a **maximal depth of 10**, **prepruning** with a minimal gain of 0.01, and a minimal leaf size of 2. The tree learns rules from columns such as last5_avg_runs, format_avg_before, bat_second_avg_before, prev_innings_runs and date to estimate how many runs Virat Kohli is likely to score.



Process design: Read CSV, Decision Tree, and the Decision Tree parameters.



Resulting regression tree (Graph view). Each box is a question about a column and each end box is a predicted runs value.

Summary: Virat Kohli's ODI data was split about 70/30 (201 train, 86 test), then a Decision Tree (a single regression tree) was trained to predict runs.